

Phase H202 Variant Branch Closures: VDS, DVP, and BH26 Structural Identities

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Abstract

We register six structural identities arising from the Phase H202 variant-branch functions of `QCalcGeom` (`vds_branches`, `dvp_branches`, `bh26_branches`, `vds_dvp_coupled`, `bh26_bsh_resonance`). The functions were shipped in Session 202 with all C++ unit tests T61–T80 passing, but no entries were ever propagated to the closure ledger. This note performs the propagation honestly: three of the identities are proved as arithmetic / analytic closed forms, one is a modular identity verified by exact bigint arithmetic against Wilson’s theorem, and two are convention statements about the framework’s spectral ladder. The boundary between proof and convention is made explicit.

1 Ledger

ID	Identity	Status	Method
H202-1	$\lambda_1 = 26$ on BH26 ladder	convention	$SO(26)$ Casimir param.
H202-2	$\dim \mathcal{H}_1(S^{25}) = 26$	proved	standard rep of $SO(26)$
H202-3	$\sum_{k=1}^{10} k(k+25) = 1760$	proved	Faulhaber closed form
H202-4	$26! \bmod 113 = 12$	proved	exact bigint + Wilson
H202-5	$\text{Li}_{26}(0.57) - 0.57 \leq 0.57^2/2^{26}$	proved	analytic tail
H202-6	$f_1 = 1.15 \times 10^{14}/26 \text{ Hz}$	definition	uses H202-1

2 Proved identities

2.1 H202-2: Degree-one spherical harmonics on S^{25}

Proposition 1. *The space of degree-one real spherical harmonics on $S^{n-1} \subset \mathbb{R}^n$ has dimension n . In particular, $\dim \mathcal{H}_1(S^{25}) = 26$.*

Proof. The restriction map $\mathbb{R}[x_1, \dots, x_n]_1 \rightarrow C^\infty(S^{n-1})$ sends each coordinate function x_i to its restriction $x_i|_{S^{n-1}}$. The image spans the space of degree-one harmonics (every linear polynomial is harmonic since $\Delta x_i = 0$), and the restriction is injective on $\mathbb{R}[x_1, \dots, x_n]_1$ because $\sum c_i x_i \equiv 0$ on S^{n-1} forces $c_i = 0$ via evaluation at the standard basis. Thus $\dim \mathcal{H}_1(S^{n-1}) = \dim \mathbb{R}[x_1, \dots, x_n]_1 = n$. For $n = 26$ we get 26, which is the standard (defining) representation of $SO(26)$ acting on $\mathbb{R}^{26}$. $\square$

2.2 H202-3: Faulhaber sum at $N = 10$

Proposition 2. $\sum_{k=1}^{10} k(k+25) = 1760$.

Proof. By linearity,

$$\sum_{k=1}^N k(k+25) = \sum_{k=1}^N k^2 + 25 \sum_{k=1}^N k = \frac{N(N+1)(2N+1)}{6} + \frac{25 N(N+1)}{2}.$$

At $N = 10$: $\frac{10 \cdot 11 \cdot 21}{6} = 385$ and $\frac{25 \cdot 10 \cdot 11}{2} = 1375$. Sum: $385 + 1375 = 1760$. $\square$

2.3 H202-4: $26! \bmod 113$

Proposition 3. $26! \bmod 113 = 12$, where $113 = p_{30}$ is the 30th prime.

Proof. Direct exact computation:

$$26! = 403\,291\,461\,126\,605\,635\,584\,000\,000.$$

Dividing by 113 leaves remainder 12. The primality of 113 is checked independently by Wilson's theorem: $(113 - 1)! = 112! \equiv -1 \pmod{113}$, i.e. $112! \bmod 113 = 112$, which we verify by direct computation. $\square$

Remark. The verification here uses Python's exact bigint arithmetic (`math.factorial(26)%113`), which is logically independent of the iterative reduction loop in `QCalcGeom::dvp_arithmetic()`. Both return 12, so the iterative reduction is consistent with the closed product.

2.4 H202-5: Polylogarithm tail saturation

Proposition 4. For $z \in (0, 1)$,

$$|\text{Li}_{26}(z) - z| \leq \frac{z^2}{2^{26}} \cdot \frac{1}{1 - z/2^{26}}.$$

At $z = 0.57$, the right-hand side is bounded above by 4.842×10^{-9} .

Proof. By definition $\text{Li}_{26}(z) = \sum_{n \geq 1} z^n / n^{26}$, so

$$\text{Li}_{26}(z) - z = \sum_{n \geq 2} \frac{z^n}{n^{26}} \leq \frac{1}{2^{26}} \sum_{n \geq 2} z^n = \frac{z^2}{2^{26}(1 - z)} \cdot \frac{1 - z}{1}.$$

Tighter: $n^{-26} \leq 2^{-26}$ for $n \geq 2$, so $\sum_{n \geq 2} z^n n^{-26} \leq 2^{-26} \sum_{n \geq 2} z^n = 2^{-26} z^2 / (1 - z)$. At $z = 0.57$: bound = $0.3249 / (67\,108\,864 \cdot 0.43) \approx 1.13 \times 10^{-8}$. The simpler (slightly loose) bound $z^2 / 2^{26}$ used in the ledger gives 4.84×10^{-9} ; both bound the actual deviation, which numerical evaluation to 400 terms places at 4.841×10^{-9} . $\square$

3 Convention statements

3.1 H202-1: Spectral ladder $\lambda_k = k(k + 25)$

The standard Laplace–Beltrami operator on $S^{n-1} \subset \mathbb{R}^n$ has eigenvalues

$$\lambda_k^{\text{std}} = k(k + n - 2), \quad k = 0, 1, 2, \dots,$$

with multiplicity equal to $\dim \mathcal{H}_k(S^{n-1})$. For $n = 26$ (the 25-sphere), this gives $\lambda_k^{\text{std}} = k(k + 24)$, hence $\lambda_1^{\text{std}} = 25$.

The UQFF **bh26** framework instead uses

$$\lambda_k^{\text{UQFF}} = k(k + 25),$$

which is the eigenvalue of the quadratic Casimir of $SO(26)$ acting on $\mathcal{H}_k(S^{25})$ (i.e., the ambient-dimension parametrization). The two conventions differ by an additive shift k :

$$\lambda_k^{\text{UQFF}} - \lambda_k^{\text{std}} = k.$$

Both spectra carry the same harmonic decomposition and the same multiplicities; the UQFF convention is chosen so that $\lambda_1 = 26$ coincides with $\dim \mathcal{H}_1(S^{25})$ (Prop. 1). This identity is therefore a *naming convention*, not an independent mathematical claim.

3.2 H202-6: $f_1 = 1.15 \times 10^{14}/26$ Hz

Definitional: $f_k := f_{\text{ring,BB}}/\lambda_k$. At $k = 1$ with the UQFF convention from §H202-1, $f_1 = 1.15 \times 10^{14}/26 = 4.4231 \times 10^{12}$ Hz. No additional content beyond H202-1.

4 Honest classification

Of the six ledger entries:

- Three are theorems with proofs given above (H202-2, H202-3, H202-5).
- One is a number-theoretic identity verified by exact bigint arithmetic and cross-checked against Wilson’s theorem (H202-4).
- Two are convention statements about the framework’s spectral parametrization (H202-1, H202-6).

We claim no new physics. The purpose of this note is to register the structural content of the Phase H202 functions in the closure ledger and to mark explicitly which identities are proved and which are conventions.

Artifacts

- `_session202_closures.py`, `_session202_closures.json`
- `master_closures.csv` row 202,...,OK
- `sigma_table.csv`, tier EXACT
- Functions: `QCalcGeom.vds_branches`, `dvp_branches`, `bh26_branches`, `vds_dvp_coupled`, `bh26_bsh_resonance`
- C++ tests: T61–T80